

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0605

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

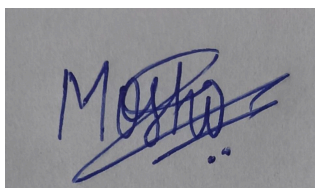
Annexure A

Case Study

Code of Conduct:

*I Moshika M - 192424064 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

A handwritten signature in blue ink, appearing to read 'Moshika', is written over a grey rectangular background.

Annexure A

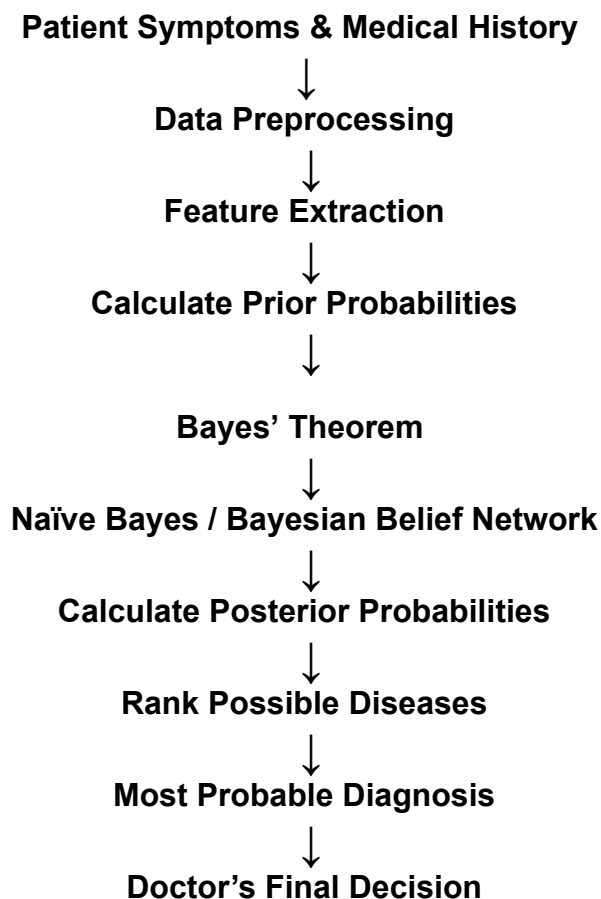
Short Answer Test

1. A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

ANSWER:

A hospital can develop an intelligent disease diagnosis system using Bayes' Theorem, Naïve Bayes, and Bayesian Belief Networks (BBNs). The system takes patient symptoms, medical history, test results, age, and other relevant information as input and calculates the probability of different diseases. Since several diseases can produce similar symptoms, a probabilistic approach is more suitable than a simple rule-based system.

FLOWCHART:



The main objective is to assist doctors by identifying the most probable diseases based on available evidence. The system does not provide an absolute diagnosis; instead, it ranks possible diseases according to their probabilities.

Bayes' Theorem is the foundation of the proposed system. It is used to update the probability of a disease when new evidence is available.

$$P(\text{Disease} \mid \text{Evidence}) = [P(\text{Evidence} \mid \text{Disease}) \times P(\text{Disease})] / P(\text{Evidence})$$

Here, $P(\text{Disease})$ is the prior probability, $P(\text{Evidence} \mid \text{Disease})$ is the likelihood of observing the symptoms, and $P(\text{Disease} \mid \text{Evidence})$ is the posterior probability after considering the patient evidence.

For example, suppose a patient has fever, cough, and fatigue. The system can calculate the probability of diseases such as influenza, pneumonia, and bronchitis. If the calculated probabilities are 60%, 30%, and 10% respectively, influenza becomes the most probable diagnosis. However, the doctor can request additional tests before confirming it.

Naïve Bayes can be used for efficient classification of patients. It assumes that symptoms are conditionally independent when the disease is known. The probability can be calculated as:

$$P(D \mid S_1, S_2, \dots, S_n) \propto P(D) \times P(S_1 \mid D) \times P(S_2 \mid D) \times \dots \times P(S_n \mid D)$$

This method is simple and computationally efficient, but the independence assumption may not always be realistic. For example, fever and fatigue may both be caused by an underlying infection and are therefore related.

A Bayesian Belief Network can overcome this limitation by representing conditional dependencies between variables. Diseases, symptoms, and test results can be represented as nodes, while their relationships are represented using directed connections.

Disease $\rightarrow$ Infection $\rightarrow$ Fever
Disease $\rightarrow$ Respiratory Problem $\rightarrow$ Cough
Disease $\rightarrow$ Infection $\rightarrow$ Fatigue

This allows the system to consider how one medical condition influences multiple symptoms. Therefore, BBNs can provide a more realistic representation of medical relationships than Naïve Bayes.

Uncertainty is handled through probability. A patient may not have all the symptoms associated with a disease, and the same symptom may occur in several diseases. Instead of making a definite decision, the system produces probability values. This allows doctors to understand the confidence of the prediction and consider additional evidence.

Probability learning is performed using historical medical records. The system learns how frequently diseases occur and how strongly particular symptoms are associated with them. A larger and more diverse dataset generally produces more reliable probability estimates. Techniques such as Laplace smoothing can also be used when certain symptom-disease combinations are missing from the training data.

The system can be evaluated using accuracy, precision, recall, F1-score, sensitivity, specificity, and a confusion matrix. In medical diagnosis, recall is particularly important because failing to identify a patient with a serious disease can have significant consequences.

The major real-world challenges include missing medical records, noisy data, rare diseases, biased datasets, and changing medical conditions. Patient privacy and security must also be maintained. Another important limitation is that the system should not replace doctors because final diagnosis requires clinical expertise.

Overall, the combination of Bayes' Theorem, Naïve Bayes, and Bayesian Belief Networks provides an effective probabilistic framework for disease diagnosis. It can learn from historical data, handle uncertainty, and represent relationships between symptoms. When properly validated and used as a decision-support tool, the system can improve the efficiency and reliability of medical diagnosis.

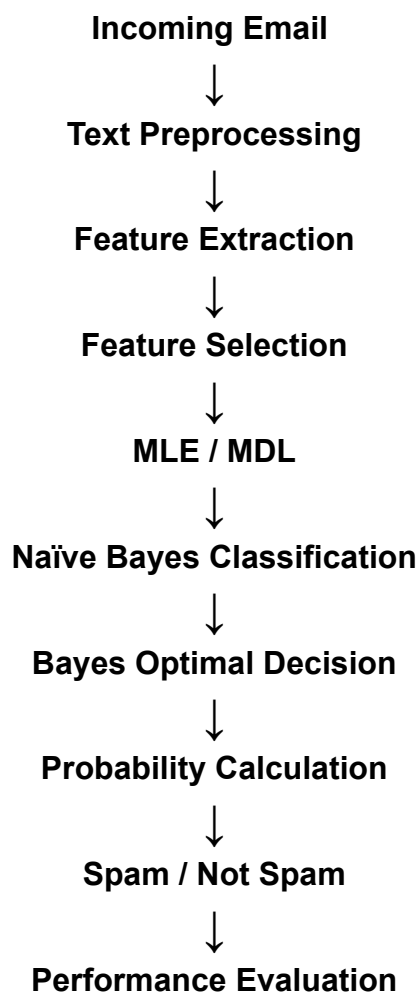
2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

ANSWER:

An organization can develop an automated spam detection system using Naïve Bayes and the Bayes Optimal Classifier. The system learns from previously labelled emails and predicts whether a new email is spam or legitimate. This reduces manual filtering and helps protect users from unwanted or potentially harmful messages.

The system can use features such as keywords, word frequency, sender information, suspicious URLs, attachments, email length, capitalization, and special characters.

FLOWCHART:



Naïve Bayes calculates the probability of an email belonging to each class. The basic relationship is:

$$P(\text{Class} \mid \text{Features}) \propto P(\text{Class}) \times P(\text{Features} \mid \text{Class})$$

The system calculates the probability of both Spam and Not Spam. If an email contains words such as "free", "winner", "offer", and "prize", and these words frequently occur in spam emails in the training dataset, the probability of spam will increase.

For example, consider an email saying:

"Congratulations! You have won a free prize. Click here to claim your reward."

The system extracts features such as "won", "free", "prize", "click", and "reward". If these features are strongly associated with spam, the system may produce:

$$P(\text{Spam} \mid \text{Features}) = 0.95$$

$$P(\text{Not Spam} \mid \text{Features}) = 0.05$$

Therefore, the email is classified as spam.

The Bayes Optimal Classifier considers multiple possible hypotheses instead of relying on only one hypothesis. It selects the classification with the highest expected probability. This can improve decision-making when several possible models can explain the training data, although it may require greater computational resources.

Maximum Likelihood Estimation (MLE) estimates model parameters by selecting values that make the observed training data most probable. In spam detection, MLE can be used to estimate the probability of words occurring in spam and legitimate emails. It works effectively when sufficient training data is available.

The Minimum Description Length (MDL) principle focuses on selecting a model that provides a good explanation of the data while keeping the model as simple as possible. It minimizes the combined description length of the model and the data. Therefore, MDL can remove unnecessary or redundant features.

The main difference is that MLE focuses on fitting the observed data, while MDL considers both model fit and complexity. A model with too many features may fit training data very well but can overfit. MDL helps control this problem by preferring simpler models.

Sample size also affects classification accuracy. With a small number of training emails, probability estimates may be unreliable. A larger dataset provides more examples of different types of spam and legitimate emails, allowing the classifier to learn more accurate patterns.

The hypothesis space also affects performance. A large hypothesis space allows the system to represent more complex patterns but increases the possibility of overfitting. A small hypothesis space is simpler and faster but may not capture sophisticated spam patterns. Therefore, an appropriate balance is required.

Performance can be evaluated using accuracy, precision, recall, F1-score, and false-positive rate. Precision is particularly important because legitimate emails incorrectly classified as spam may cause important messages to be missed. Recall measures how effectively the system identifies actual spam.

Real-world spam detection also faces changing spam patterns, new vocabulary, phishing techniques, and class imbalance. Therefore, the model should be retrained regularly using updated email data

Thus, Naïve Bayes provides a fast and practical method for spam classification, while the Bayes Optimal Classifier provides a broader probabilistic decision framework. MLE is useful for probability estimation, whereas MDL helps control model complexity and select useful features. A sufficiently large dataset and suitable hypothesis space are essential for achieving good classification accuracy.

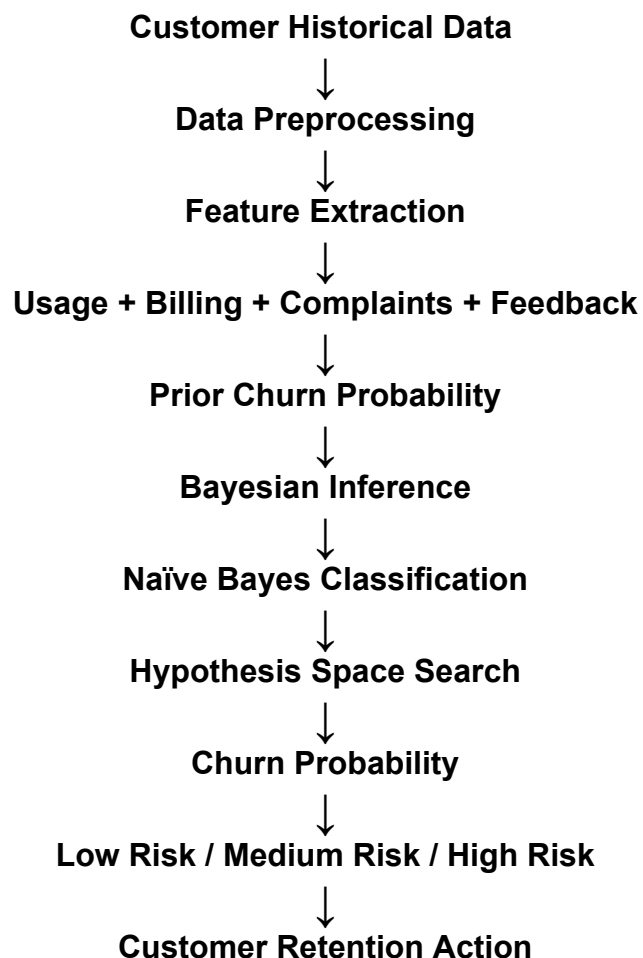
3. A telecommunications company aims to predict customer churn using historical usage data and service feedback. Using probabilistic models such as Naïve Bayes, Bayesian inference, and hypothesis space search, design a predictive system. Analyze how prior probabilities and feature dependencies influence churn prediction. Evaluate how uncertainty in customer behavior is handled and discuss the effectiveness of the model in improving customer retention strategies.

ANSWER:

A telecommunications company can develop a customer churn prediction system using Naïve Bayes, Bayesian inference, and hypothesis space search. Customer churn occurs when customers stop using a company's services or move to a competitor. Predicting churn early allows the company to take preventive action and improve customer retention.

The system can use customer information such as call duration, internet usage, monthly bill, contract type, complaints, payment history, service quality, and customer feedback.

FLOWCHART:



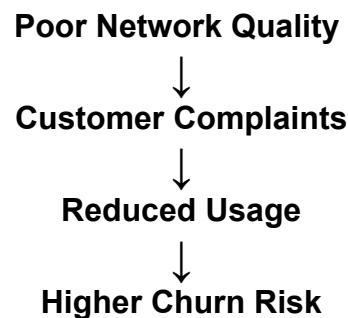
Naïve Bayes can classify customers into Churn and No Churn categories. The probability is calculated using:

$$P(\text{Churn} \mid \text{Features}) \propto P(\text{Churn}) \times P(\text{Features} \mid \text{Churn})$$

The prior probability represents the historical probability of customers leaving the company. For example, if 15% of customers previously churned, the initial probability of churn is 0.15.

When new evidence is observed, Bayesian inference updates this probability. For example, a customer who initially has a low churn probability may start making complaints, reduce their internet usage, and experience poor service quality. These new observations can increase the posterior probability of churn.

Feature dependencies are important because customer characteristics are often related. For example:



Naïve Bayes assumes that features are conditionally independent, but in reality, poor network quality may cause both complaints and reduced usage. Ignoring such dependencies can sometimes lead to inaccurate probability estimates. More advanced Bayesian models can represent these relationships explicitly.

Hypothesis space search allows the system to consider different possible relationships between customer features and churn. For example, one hypothesis may suggest that high billing increases churn, while another may suggest that poor network quality combined with high billing is a stronger indicator. The system can compare these hypotheses using historical data.

Uncertainty is handled by producing probabilities instead of definite predictions. For example:

Customer A – 20% churn probability – Low Risk

Customer B – 55% churn probability – Medium Risk

Customer C – 85% churn probability – High Risk

This allows the company to focus its retention efforts on customers with the highest risk.

Consider a customer with high monthly billing, decreasing internet usage, several complaints, poor service quality, and a contract close to expiration. Based on historical data, the model may calculate:

$$P(\text{Churn} \mid \text{Customer Features}) = 0.82$$

The customer can therefore be classified as high risk. The company can contact the customer and provide technical support, a suitable plan, or a personalized offer.

The performance of the model can be evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix. Recall is important because the company wants to identify as many potential churners as possible. However, precision is also important because unnecessary discounts or offers can increase business costs.

The predictions can directly support customer retention. High-risk customers can receive personalized discounts, improved plans, technical assistance, loyalty benefits, or targeted communication. This makes the retention strategy more focused instead of offering the same solution to every customer.

Real-world challenges include incomplete customer data, changing customer preferences, class imbalance, inaccurate feedback, and external factors such as competitor offers. Customer privacy must also be protected because the system processes behavioural and personal information.

Therefore, a probabilistic churn prediction system can help telecommunications companies identify high-risk customers before they leave. Naïve Bayes provides an efficient classification method, Bayesian inference allows predictions to be updated as new evidence becomes available, and hypothesis space search helps identify useful relationships between customer features. By handling uncertainty and using historical patterns, the system can support targeted and effective customer retention strategies.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand